OPTIMAL CODES AND ARCS FOR THE GENERALIZED HAMMING WEIGHT

Assia Rousseva
Sofia University

(joint work with Sascha Kurz, Ivan Landjev)

– WCC 2026, Paris, France, June 8–12, 2026 –

1. Linear Codes

A linear $[n, k]_q$ code C is a k-dimensional subspace of $\mathbb{F}_q^n$.

For $c = (c_1, \ldots, c_n) \in \mathbb{F}_q^n$: $\mathrm{Supp}(c) := \{1 \leq i \leq n \,:\, c_i \neq 0\}$ the support of c and $\mathrm{wt}(c) = |\,\mathrm{Supp}(c)|$ its weight (Hamming weight).

$$\mathrm{Supp}(C) := \{1 \leq i \leq n \,:\, \exists c = (c_1, \ldots, c_n) \in C, c_i \neq 0\}\,.$$

The rth generalized Hamming weight of a linear code C denoted as $d_r(C)$, is the minimal support size of an r-dimensional subcode of C:

$$d_r(C) := \min\{|\,\mathrm{Supp}(C')| \,:\, C' \leq C, \dim(C') = r\}\,.$$

In particular, $d_1(C)$ is the minimum Hamming distance of the code C.

A linear code of length n, dimension k and rth generalized Hamming weight equal to d_r: $[n, k, d_r]^{(r)}$-code.

The sequence $(d_1(C), \ldots, d_k(C))$ is called the weight hierarchy of a linear $[n, k]_q$ code C.

It holds,

$$1 \leq d_1(C) < d_2(C) < \cdots < d_k(C) \leq n.$$

2. Arcs and Minihypers

A multiset in $\mathrm{PG}(k-1,q)$ is a mapping $\mathcal{K}\colon \mathcal{P} \to \mathbb{N}_0$.

This mapping is extended to any subset $\mathcal{Q}$ of $\mathcal{P}$ by $\mathcal{K}(\mathcal{Q}) := \sum_{P\in\mathcal{Q}} \mathcal{K}(P)$.

- $w_r := \max\left\{\mathcal{K}(S)\ : S \text{ is a subspace of } \mathrm{PG}(k-1,q), \dim(S) = r\right\};$
- $u_r := \min\left\{\mathcal{K}(S)\ : S \text{ is a subspace of } \mathrm{PG}(k-1,q), \dim(S) = r\right\}.$
- $(n, w_r)^{(r)}$-arc: if each r-dimensional subspace of $\mathrm{PG}(k-1,q)$ has multiplicity at most w_r and there exists an r-dimensional subspace with this multiplicity;
- $(n, u_r)^{(r)}$-minihyper: if each r-dimensional subspace of $\mathrm{PG}(k-1,q)$ has multiplicity at least u_r, and there exists an r-dimensional subspace with this multiplicity.

3. Equivalence of Codes and Arcs

Let C be a linear code of full length ($d_k(C) = n$) with parameters $[n, k]_q$.

Let $G = (g_1^T \cdots g_n^T)$, $g_i \in \mathbb{F}_q^k$, be a generator matrix of C.

The columns g_i^T are considered as points P_i given by their homogeneous coordinates in $\mathrm{PG}(k-1, q)$.

The generator matrix G is associated with an ordered n-tuple of points $(P_1, \ldots, P_n)$ in $\mathrm{PG}(k-1, q)$.

This n-tuple defines a multiset $\mathcal{K}$ by $\mathcal{K}(P) := |\{i | P_i = P\}|$.

Theorem 1. Let C be an $[n, k]_q$-code of full length and let $\mathcal{K}$ be the arc in $\mathrm{PG}(k-1, q)$ associated with C. Then:

$$w_r(\mathcal{K}) + d_{k-r-1}(C) = n.$$

for every $r = 0, \cdots, k-1$

Theorem 1 implies

$$1 \leq w_0 < w_1 < \cdots < w_{k-1} = n.$$

If $s \geq w_0$ is an integer, then $\mathcal{F} = s\chi_{\mathcal{P}} - \mathcal{K}$ is a minihyper of cardinality $sv_k - n$ with $u_r = sv_{r+1} - w_r$, $r = 0, 1, \ldots, k-1$.

Here $v_r = (q^r - 1)/(q-1)$.

4. The Main Problem for Linear Codes, Arcs and Minihypers

- $m_q^{(r)}(k-1,w)$ – the maximal cardinality of a multiset in $\mathrm{PG}(k-1,q)$ such that every r-dimensional subspace has multiplicity at most w.

  In other words, $m_q^{(r)}(k-1,w)$ is defined as the largest n for which there exists an $(n,w)^{(r)}$-arc in $\mathrm{PG}(k-1,q)$.

- $n_q^{(r)}(k,d)$ – the smallest length for which there exists an $[n,k,d]_q^{(r)}$-code.

Main problem:

(1) Finite geometry: Given the integers w, k, q, find $m_q^{(r)}(k-1, w)$.

(2) Coding theory: Given the integers d, k, q, find $n_q^{(r)}(k, d)$.

Some special cases:

- $r = 0$: $m_q^{(0)}(k-1, s) = sv_k$ (trivial)
- $r = 1$: $m_q^{(1)}(k-1, w)$ – the largest size of a generalized cap

  In particular: for $w = 2$ – this is the largest cap problem

$$m_q^{(1)}(2,2) = \begin{cases} q+1 & q \text{ odd}, \\ q+2 & q \text{ even}. \end{cases} \quad m_q^{(1)}(3,2) = q^2 + 1.$$

$m_3^{(1)}(4,2) = 20$, (G. Pellegrino)

$m_3^{(1)}(5,2) = 56$, (R. Hill)

$m_3^{(1)}(6,2) = 112$, (A. Potechin)

$m_4^{(1)}(4,2) = 41$. (G. Tallini; Y. Edel, J. Bierbrauer)

- $r = k - 2$: the classical largest arc problem

5. The Griesmer Bound for the Genralized Hamming Weights

The Griesmer bound (Griesmer, 1960):

Theorem 2. If C is a linear $[n,k,d]_q$-code then

$$n \geq \sum_{i=0}^{k-1} \left\lceil \frac{d}{q^i} \right\rceil =: g_q(k,d).$$

Griesmer code: An $[n,k,d]_q$ code with $n = g_q(k,d)$.

The generalized Griesmer bound (T. Helleseth, T. Kløve, Ø. Ytrehus, 1992):

Theorem 3. Let C be an $[n,k,d_r]^{(r)}$-code, where $1 \leq r \leq k$. Then

$$n \geq d_r + \sum_{i=1}^{k-r} \left\lceil \frac{d_r}{q^i v_r} \right\rceil =: g_q^{(r)}(k,d).$$

Here $v_r = \dfrac{q^r - 1}{q - 1}$.

6. Results for the Optimal Codes and Arcs

Let d be a positive integer. It can be written uniquely as

$$d = sq^{k-1} - \sum_{i=0}^{k-2} \varepsilon_i q^i \tag{1}$$

$$g_q^{(1)}(k,d) = sv_k - \sum_{i=0}^{k-2} \varepsilon_i v_{i+1}.$$

Let C be an $[n = g_q^{(1)}(k,d), k, d]_q^{(1)}$-code and let $\mathcal{K}$ the $(n, n-d)$-arc in $\mathrm{PG}(k-1,q)$ associated with C.

$$w_{k-1-j} = s v_{k-j} - \sum_{i=j}^{k-2} \varepsilon_i v_{i+1-j}.$$

Then $\mathcal{K}$ can be represented as

$$\mathcal{K} = s\chi_{\mathcal{P}} - \mathcal{F},$$

where $\mathcal{F}$ is a minihyper with parameters

$$\left(\sum_{i=0}^{k-2} \varepsilon_i v_{i+1}, \sum_{i=0}^{k-2} \varepsilon_i v_i \right)$$

with maximal multiplicity of a point not exceeding s.

Minihypers with the above parameters, but without the restriction on the maximal point multiplicity, can be constructed always as the sum of subspaces: ε_{k-2} hyperplanes, ε_{k-3} hyperlines, and so on, ε_1 lines and ε_0 points. Such minihypers are called canonical.

Minihypers with parameters

$$\left(\sum_{i=r}^{k-2} \varepsilon_i v_{i+1}, \sum_{i=r}^{k-2} \varepsilon_i v_{i+1-r}\right)^{(k-1-r)} \tag{2}$$

can be constructed in a similar fashion as canonical minihypers, i.e. as the sum of ε_{k-2} hyperplanes, ε_{k-3} hyperlines, and so on, ε_r subspaces of dimension r.

Theorem 4. Let $\mathcal{F}$ be a minihyper in $\mathrm{PG}(k-1,q)$ with parameters given by (2), $0 \leq \varepsilon_i \leq q-1$, and with maximal point multiplicity w_0. Then for every $s \geq w_0$ the multiset $\mathcal{K} = s\chi_{\mathcal{P}} - \mathcal{F}$ is an arc in $\mathrm{PG}(k-1,q)$ with parameters

$$\left(n = sv_k - \sum_{i=r}^{k-2} \varepsilon_i v_{i+1}, w_{k-r-1} = sv_{k-r} - \sum_{i=r}^{k-2} \varepsilon_i v_{i+1-r}\right)^{(k-1-r)}.$$

The code associated with $\mathcal{K}$ is a Griesmer code with respect to the r-th generalized Hamming weight.

Theorem 5. If d_r is large enough Griesmer $[n = g_q^{(r)}(k, d_r), k, d_r]_q^{(r)}$-codes do exist for all q, k and r.

Theorem 6. Let C be an $[n = g_q^{(1)}(k, d), k, d]_q^{(1)}$-code where $d^{(1)} = d$ is given by (1). Then

$$d_r(C) = v_r \cdot \left(sq^{k-r} - \sum_{i=r}^{k-1} \varepsilon_{i-1} q^{i-r} \right) - \sum_{i=1}^{r-1} \varepsilon_{i-1} v_i,$$

and C attains the Griesmer bound for the r-th generalized Hamming weight, i.e. $n = g_q^{(r)}(k, d_r)$.

Theorem 7. Every $(xv_{k-1}, xv_r)^{(r)}$-minihyper in $\mathrm{PG}(k-1,q)$, $q = p^h$, $k \geq 4$, with $x \leq q - \frac{q}{p}$ is a sum of hyperplanes for all $r \in \{1, \ldots, k-2\}$.

Theorem 8. Every $(xv_{k-1} + v_{k-2}, xv_r + v_{r-1})^{(r)}$ in $\mathrm{PG}(k-1,p)$, p prime, for $x \leq p - 2$ is the sum of x hyperplanes and one subspace of codimension 2.